

Scheduler — Black-box I/O

One channel, DDR5. Every signal in/out of the scheduler (arb tree): can-bits, demand, maintenance flags, grants, bundle. All single-bit unless width noted. Parametric: $N_BANKS (=N_BG \times 4)$, N_BG , N_RANKS .

1 INPUTS

Per-bank demand legality (from cells / timestamp scoreboard)

signal	width	from/to	meaning
can_act[b]	$1 \times N_BANKS$	in	ACT legal: $GC \geq next_act$ & state ok
can_cas[b]	$1 \times N_BANKS$	in	CAS legal: $GC \geq next_cas$ & state==OPEN & row_hit
can_pre[b]	$1 \times N_BANKS$	in	PRE legal: $GC \geq next_pre$
need_act / need_cas / need_pre[b]	$1\ ea \times N_BANKS$	in	what the resident packet wants (row miss->pre/act, hit->cas)
is_hit[b]	$1 \times N_BANKS$	in	open_row == pkt row (feeds CAS_W weight)
row_valid[b]	$1 \times N_BANKS$	in	OPEN & !pre_pending (admit window)
age[b]	few $\times N_BANKS$	in	age-of-need (feeds ACT_W, capped)

Cross-scope legality

signal	width	from/to	meaning
can_act_bgs[g] / can_cas_bgs[g]	$1\ ea \times N_BG$	in	same-BG spacing met ($tRRD_L$ / $tCCD_L(_WR)$)
can_act_bgd[r] / can_act_rank[r]	$1\ ea \times N_RANKS$	in	cross-BG ACT ($tRRD_S$) + 4-ACT window ($tFAW$)
can_wr[r] / can_rd[r]	$1\ ea \times N_RANKS$	in	$R < \rightarrow W$ turnaround gap met ($tWTR$ / $tRTW$)
can_cas_dq	1	in	DQ bus free ($dqFree=BL/2$); cross-rank tRTRS folded

Maintenance / refresh flags (from ME)

signal	width	from/to	meaning
o_ref_rdy	1	in	refresh owed (!debt) – issue if target idle
ref_force	1	in	forced refresh (&debt==7)
rfm_rdy / rfm_force	$1\ ea$	in	rowhammer owed ($RAA \geq IMT$) / hard ($\geq MMT$)
zq_rdy / mrw_rdy / mrr_rdy	$1\ ea$	in	ZQ / MRW / MRR ready
cas_complete[b]	$1 \times N_BANKS$	in	bank drained + precharged (state==IDLE)
maint_cmd_type	5	in	MAINT_PRIO winner (REF/RFM/ZQ/MRW/MRR)
init_done	1	in	boot complete (else scheduler gated, DFI mux on init)

Control / bus state

signal	width	from/to	meaning
batch_dir[r]	$1 \times N_RANKS$	in	WLR read/write batch direction
last_cas_dir / rank / bg	few	in	bus state (turnaround + diff-BG packing decisions)
gear	2	in	mc:CK ratio (bundle width = GEAR phases)
req_valid (per lane)	$1 \times N_RANKS$	in	lane has an admitted packet (from lookahead)
dfi_alert_n / dfi_error	$1\ ea$	in	PHY alert / error (retry / stub)

2 OUTPUTS

Grants (to cells)

signal	width	from/to	meaning
grant_cas_oh[b]	$1 \times N_BANKS$	out	CAS winner one-hot; pop-on-CAS (retire packet)
grant_act_oh[b]	$1 \times N_BANKS$	out	ACT winner one-hot; commits ACTING
grant_pre_oh[b]	$1 \times N_BANKS$	out	PRE winner(s) one-hot (≤ 2); commits PREING

Phaser bundle (to phaser / DFI encode)

signal	width	from/to	meaning
record[k]	$\sim 49 \times K(\leq 3)$	out	{cmd_type[5], rw, rank, bg, bank, addr[18], ap, bl, rob_index[8], sram_addr[9]}
placement[pN]	{cmd_sel[2], ui} $\times$ GEAR	out	per-phase: which record + which UI (2-UI cmd = 2 slots)
two_ui / repeat	1	out	separate side wire -> DFI driver (1-UI vs 2-UI; @1:1 hold)

Commit / feedback

signal	width	from/to	meaning
fsm_commit (issue-edge)	pulse $\times$ class	out	sole commit: updates BANK_FSM state + fires timestamp writeback
cmd_issued{r,bg,bank,cmd}	bus	out	drives the next_* writeback adders (WE + const-router select)
pop_cas	1	out	retire the CAS lane's packet (BANK_REQ free)

Maintenance handshake (to ME)

signal	width	from/to	meaning
<code>maint_ack / maint_pop</code>	1	out	the maint cmd (<code>cmd_type</code>) was issued -> ME clears debt/RAA, advances ba
<code>stall_acts drive</code>	(ME-owned)	—	NOTE: <code>stall_acts</code> is set by ME (drain req), not the scheduler; scheduler only ranks <code>ref_rdy/force</code> with demand

How it works (one line): the scheduler NEVER rejects — all legality is folded into the `can_*` inputs, so the arb only SELECTS. Demand lanes (`cas/act/pre`) rank per-bank weights (`CAS_W` hit-lock, `ACT_W` age, `PRE_W` close-row) up through BGB/BG/L3 into an N-winner bundle; maintenance enters as flags (`o_ref_rdy/force`, `rfm_*`, `zq/mrw/mrr_rdy`) ranked WITH demand (`MAINT_PRIO`), override-injected at the phaser. Outputs = per-class `grant_oh` (pop-on-CAS) + the phaser bundle + the `fsm_commit` pulse (sole state+timestamp writeback edge).